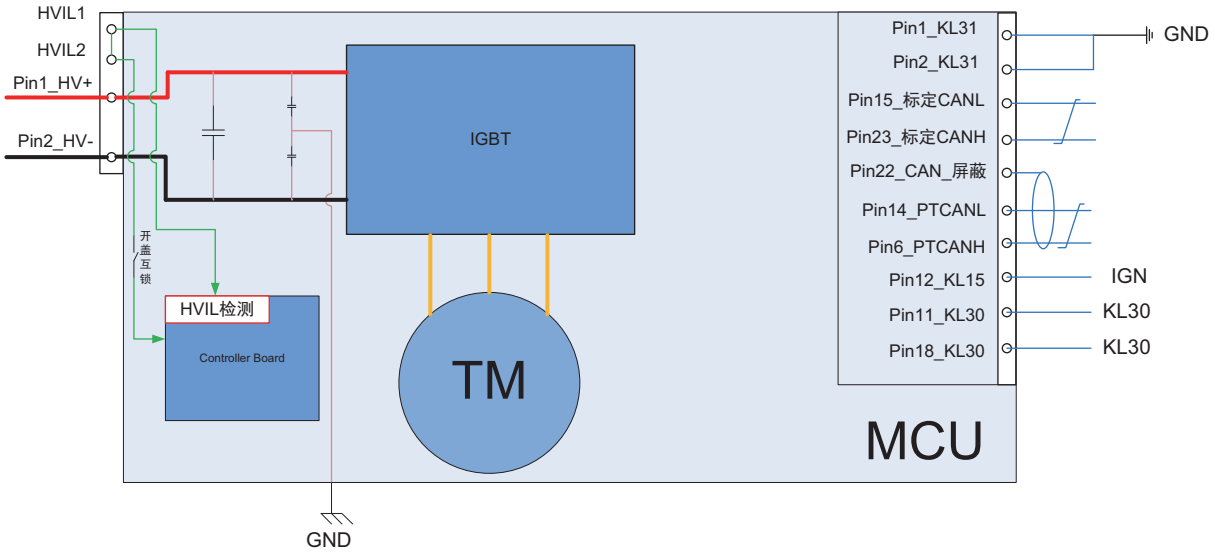
## P0A90-00

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|---|
| **Fault diagnosis code** |
| P0A90-00：Motor stall fault |
| **Detection tools** |
| 1 Multimeter and diagnostic apparatus |
| **Possible causes** |
| • Circuit,terminal and adapting piece fault<br>• PEU module fault<br>• Driving motor fault<br>• PEU motor monitoring circuit fault |
| **Conditions for fault code operation** |
| Turn the ignition switch to ON position. |
| **Conditions for fault code setting** |
| P0A90-00 |
| The test starts when the speed is lower than 50rpm and the torque is higher than 180N.m. The limit peak torque reaches 270N.m after 1s. The speed is higher than 150rpm or the torque is lower than 150N.m for 0.5s, and the exit is instant. The total duration of the stalled rotor is reported after 15s. |
| **To set the effect of a fault code condition** |
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**Circuit diagram**




EV901001

| **Troubleshooting steps** | | |
|---|---|---|
| Step 1 | General inspection. | |
| | | 1 Check the fuse for damage.<br>2 Check battery capacity.<br>3 Check the harness plug of PEU for damage, poor contact, aging and looseness.<br>4 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 2 | Check fault codes. | |
| | | 1 Connect the fault diagnosis instrument to the diagnosis interface.<br>2 Turn the ignition switch to ON position.<br>3 Diagnose the PEU with diagnostic apparatus.<br>4 Is there any other fault code except for P0A90-00? |
| | | **YES →** Refer to:DTC Summary list(PEU). |
| **NO ↓** | | |
| Step 3 | Check the parking system. | |
| | | 1 Check the mechanical parts of the parking system for modification and jamming.<br>2 Test whether the fault is existed. |
| | | **YES →** Repair or replace the faulted parts. |
| **NO ↓** | | |
| Step 4 | Check the connecting cable of PEU. | |
| | | 1 Turn the ignition switch to OFF.<br>2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual.<br>3 Disconnect the harness plug battery pack high voltage.<br>4 Disconnect the high voltage harness plug of PEU.<br>5 Check whether the PEU connection cable is rotten, damaged, and water enters.<br>6 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |

| | | |
|---|---|---|
| Step 5 | Perform initialization. | |
| | | 1 Turn the ignition switch to OFF.<br>2 Connect the fault diagnosis instrument to the diagnosis interface.<br>3 Turn the ignition switch to ON.<br>4 The governor was initialized with the diagnostic instrument.<br>5 Can the governor be initialized and is the governor restored? |
| | | **NO →** Repair or change the governor. |
| **YES ↓** | | |
| Step 6 | Check the insulation of PEU. | |
| | | 1 Use a megohm meter to check the insulation resistance of the PEU housing and grounding according to the procedure in the maintenance manual.<br>**Standard value: >20MΩ**<br>2 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 7 | Check the insulation of driving motor. | |
| | | 1 Use a megohm meter to check the insulation resistance of the driving motor housing and grounding according to the procedure in the maintenance manual.<br>**Standard value: >20MΩ**<br>2 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 8 | Check PEU internal circuit. | |
| | | 1 Turn the ignition switch to OFF.<br>2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual.<br>3 Open the PEU end cover.<br>4 Check the PEU internal circuit and components for obvious damage or . ablation.<br>5 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |

| Step 9 | Test whether PEU is norm. |
|---|---|
| | 1 Turn the ignition switch to OFF.<br>2 Disconnect and check the harness plug EB1002 of PEU.<br>3 Check and repair.<br>• Corrosion(install new adapting piece or terminal-cleaning module and pin).<br>• Pin damaged or bent-install new terminal/pin.<br>4 Fly out of pins-install new pins as needed.<br>5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.<br>6 Run the system and determine if the problem persists? |
| | **YES →** Change the PEU Module |
| **NO ↓** | |
| At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems. | |
| **Next Step ↓** | |
| Diagnosis end. | |